

Taking Digital Product Design Coordination to the Next Level by Elastic Zooming and Linecept

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1 PROBLEM

Currently, we use a multitude of computing environments in our daily lives, and these devices are increasingly becoming an essential part of our everyday lives [7, 10]. For example, someone might start the day by reviewing notifications on a smartwatch, move to browsing the news and quickly responding to emails during breakfast on a smartphone, and then using a laptop and digital whiteboard during work hours to send emails and facilitate meetings. Sometimes, even multiple computing environments are used in parallel to fulfill a task — Waeljas *et al.* describe these are multi-channeled and cross-media service experiences [12]. However, because existing software programs are seldom able to provide a seamless experience across multiple devices, users are forced to “act as the bridge connecting their devices” [4]. This bridging creates additional burdens on users, which have been found to increase cognitive load [4]. Additionally, even on the same screen, we use variable viewport sizes to run multiple software applications. These phenomena are more and more observable on handheld devices as well. In digital product design, the use of multiple devices and multiple software is even more augmented as a wide variety of stakeholders with different cultural backgrounds and tool preferences need to work together in order to efficiently deliver scalable and secure digital solutions to billions of customers. Until now, there has yet to be a uniform visualization technique that can serve as a main metaphor for user interfaces across a wide variety of screen sizes and shapes to access information. Such a uniform solution would allow a user to interact with displayed information in a fully consistent way from

*David Kutas previously had financial interests in Linecept, Inc. a former company with interests related to the user study. The nature of this financial interest and the design of the study have been reviewed by the UCI Conflict of Interest Oversight Committee (COIOC). The COIOC has determined that the researcher’s previous financial interests were appropriately managed as to avoid compromising the quality or reliability of the study and furthermore, the UCI Institutional Review Board has determined that appropriate safeguards were in place to avoid adversely affecting the study participants’ safety and welfare.

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watch-sized to whiteboard-sized screens, and would also decrease the development time of digital products. As an additional benefit of a novel user interface type, it would allow the manifestation of new software productivity tools that would be significantly easier to use, which could yield to less time spent on work coordination, hence more time allowed to efficient work, and thus, leisure. This way, great societal benefits could be realized as well.

2 PROPOSED RESEARCH

The goals of my work are (1) to develop a persuasive user interface model that (2) can be used for software and product design work. The approach will provide a novel user interface model that can be applied generally in a wide variety of computing environments (from watch-sized to whiteboard-sized screens). Additionally, the approach will also provide a software productivity tool, named Linecept, that will enable software and product design practitioners as well as educators to easily understand the breadth and depth of their intermediary and final artifacts in order to better inform their work processes and to enable their instructors to quickly understand the quality of contributions by the practitioners.

3 PRIOR RESEARCH

After careful deliberation I realized that my research should be based on two pillars. The first should be a piece of infrastructure that contributes to significantly decrease the cognitive load caused by the computing environments and the circumstances of how modern digital product design is done. The other pillar being a collaborative software tool that aids to directly mitigate the discovered problems in collaborative work. In the next couple of sentences I briefly cover research that have major influence on the research I conduct.

Soon after I started my research, I acknowledged that Zoomable User Interfaces (hereinafter referred to as ZUIs) once held promise to become the main underlying interface metaphor for modern computing software. One of its main goals was to decrease cognitive load for users trying to understand complex data. Although much progress has been made to exploit ZUIs’ positive properties, this vision was largely never materialized, perhaps because we rapidly transitioned into a world where we use multiple devices at the same time. Problematically, ZUIs were not designed to fit into the substantially differing multi-device computing environments (that are manifested through significantly differing screen sizes and shapes) that we now use every day. It is remaining HCI’s “grand challenge” to exploit the full potential of ZUIs.

In their 2009 review paper [3], Cockburn *et al.* defined ZUIs to be types of user interfaces that present information with magnification metaphors and temporally separate the information. Further, Cockburn *et al.* [3] defined user interfaces based on their prevalent

design schemes. They defined four different user interface visualization mechanisms: Overview+Detail, Zooming, Focus+Context, and Cue-based. Studies have examined how people comprehend stories and presentations delivered through animated versus non-animated mediums, as well as through ZUI-based presentations versus slide-based presentations. Boltman *et al.* and Good *et al.* concluded that, while the content itself was not more easily recalled by the audience, when ZUI or rich animations were used, the structure of the content could be better recalled. These findings suggest that ZUIs may be more engaging [2, 6]. A study by Moulton [9] found similar results when they compared Prezi (ZUI-based presentation creation web application) to regular slide-based presentation formats. Study participants found the Prezi presentations to be more engaging and preferred them over regular slide presentations. Other prior and fundamental studies assessed user performance when animation was provided for a scroll action compared to when no animation was provided.

In the research and commercial fields of online product design collaboration, a number of tools (such as Google Drive, Dropbox and GitHub) and a large body of research exist. Version control systems allow users to see intermediary artifacts, such relevant visualization research tool is The Code Time Machine that allows rapid flyby of files inspired by Apple's Time Machine UI [1]. Classroom Bridge and Designer's Outpost enable capturing of intermediary artifacts such as notes and other multimedia content on user interfaces that are timeline based [5, 8]. Ben Shneiderman appointed many of the existing grand challenges [11].

4 RESEARCH PROGRESS AND FUTURE PLANS

4.1 Work to Date

First, I have developed a uniform visualization technique to access information across a wide variety of screen sizes and shapes (Video Figure 1). In service of developing such uniform visualization technique, I constructed an algorithm (named *Adaptive Magnification*). I also constructed multiple navigation methods to enable users to efficiently traverse this interface. I named this package of visualization and interaction techniques, *Elastic Zooming* and its implementation, *EZUI*. EZUIs are visual user interfaces where the prevalent interaction metaphor is based on a dual-scale factor (different scale factors are applied on the horizontal and vertical dimensions of the objects to be magnified resulting in aspect ratio change of the objects represented on the user interface) animated magnification from one visual state of the user interface to the other.

Second, I have developed a robust three-tier architecture-based web application in order to tackle the second point of my proposed research. The tool, named Linecept was deployed to 500+ students at the University of California, Irvine (UCI). The deployment revealed limitations of the research approach and the tool itself. In particular, the research goals were unclear and the software version did not have the EZUI infrastructure. Thus, I realized that before committing to a full-fledged application development, the initial vision of the infrastructure needed to be pursued, namely a user interface solution that makes it at least one magnitude easier for people to use the tool, otherwise they may simply ignore investing

effort in using the tool and return back to traditional class collaboration solutions that will not yield either to higher quality work nor interesting research insights. Additionally, I realized that a robust human-subject research plan has to be developed before using a tool in the wild.

Thus, to further evaluate the built interface with the future productivity tool in mind, I returned back to my original vision that was born prior to the start of my PhD studies. Figure 1 and 2 introduce the concept and the first representation of this novel design coordination tool.

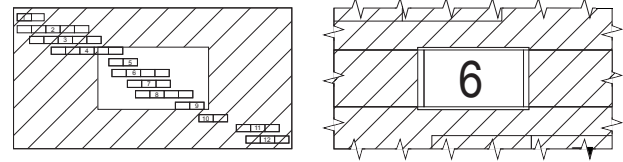


Figure 1: Conceptual schematic of the Timeline-based Multi-phase Artifact Discovery and Manipulation Tool, Linecept. The interface is in its zoomed-out state on the left and in its zoomed-in state on the right. Diagonal lines mark the areas that can be panned to, but currently fall outside the area of the viewport. Numbers 1–12 denote *Segments* of the different *Phases*. The image on the right illustrates the interface when the zooming has finished, ZL 2 is reached, and Object 6 is in the center of the viewing window. In the zoomed-in state of the user interface, Object 6 can contain *Cards*.

I have also indirectly evaluated the novel user interface model during the COVID-19 lockdown, which caused some limitations, through an Amazon Mechanical Turk crowdsourcing platform based study. The study provided a first descriptive look at participants' preferences of different interface types in case of the proposed productivity tool (Video Figure 3). A descriptive look at the results shows that the Elastic Zoomable User Interface was, overall, preferred over other interfaces. Overall, 17 out of 48 participants ranked EZUI as the interface that they preferred the most (35.42%). The achieved results are statistically non-significant, and are likely due to the present study being underpowered and not being the most preferred way to evaluate the proposed research.

4.2 Future Work

Future work will focus on building a scalable, maintainable version of the productivity tool, based on the Elastic Zoomable User Interface provided visual infrastructure. The development of the software tool itself falls outside the scope of my PhD studies at UCI. This implementation will enable me to conduct additional studies that could yield to: (1) learn more about the novel user interface, as well as to (2) discover yet hidden insights of how people collaborate together in digital spaces when they are in the creative, conceptual, less formal phases digital product design. A future empirical study will focus on further assessing the Elastic Zoom based interface in a direct interaction embodiment. I will recruit undergraduate students at UCI to test four different software prototypes in-person where they can directly interact with the prototypes, based on a provided task-oriented script. This task-oriented script would

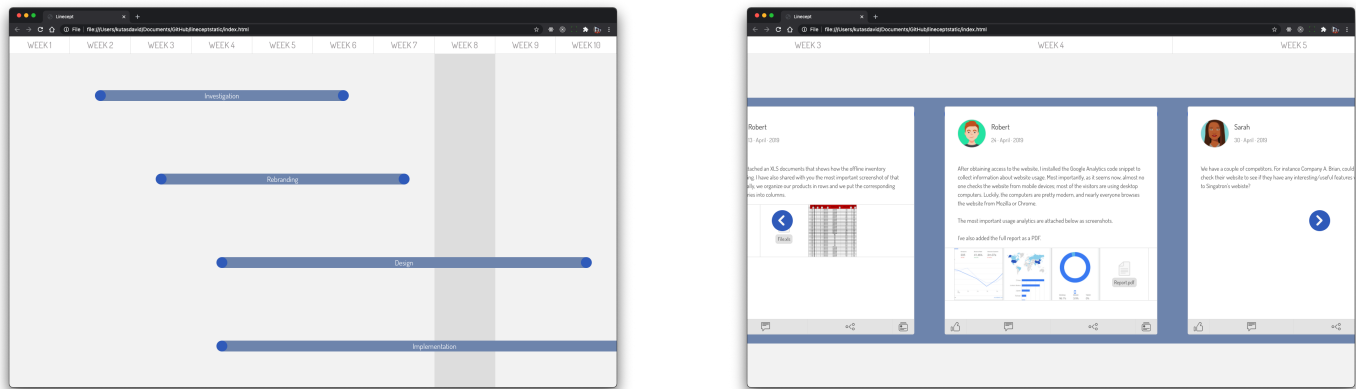


Figure 2: Actual prototype user interface of the Timeline-based Multi-phase Artifact Discovery and Manipulation Tool, Linecept. The interface is in its zoomed-out state on the left and in its zoomed-in state on the right in a desktop web browser. Video Figure 2 introduces the entire concept through a use case scenario both on a laptop and a handheld device screen.

also allow me to carry out a cognitive walkthrough. The goal of this study will be to assess the preliminary findings of the already conducted empirical study in a more realistic and comprehensive scenario. Subsequently, in another study, my goal would be to actually use Linecept for purposes that can extend the knowledge of digital product design. I will deploy Linecept in a real-life course and investigate how teams and individuals design software.

5 CONTRIBUTIONS

My dissertation when completed, will provide a number of contributions. First, it will introduce a novel user interface model that can help reduce cognitive load when people interact with complex project-related information. This could lead to positive work environment and through that societal changes. Second, it will provide a tool that can be used by industry practitioners and other researchers to further understand how people work together. Third, it will provide a set of empirical studies that will evaluate the effectiveness of the novel user interface model in contrast to other traditional graphical user interface solutions. Ultimately, in a nutshell, I wish to discover if my hypothesis about lowering the burden to understand complex projects by a highly visual and easily understandable coordination tool substantially help people or not to be more productive and less busy, hence more happy.

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A VIDEO FIGURES

A.1 Video Fig. 1. Comparison of Magnification Techniques

This video figure (file: vidFig1.mp4 attached to the submitted manuscript) is a sequence of animations that compares the one-point perspective magnification to elastic magnification. The two magnification techniques are compared on a typical laptop as well as smartphone screens. Each animation consists of an outer and inner rectangle. The inner rectangle denotes the viewing window, whereas the outer rectangle denotes the entire user interface area where all the objects after zoom would fit, and where navigation is possible, too. The navigable areas are denoted with a semi-transparent

grey color. The interactable objects (9 items) are shown with thin borders. The sequences are the following:

Part A: A summary of the comparisons. Video section: starting at 00:00.

Part B: One-point perspective magnification of objects on a typical laptop screen. Video section: starting at 00:03.

Part C: One-point perspective magnification of objects on a typical smartphone screen. Video section: starting at 00:08.

Part D: Elastic magnification of objects on a typical laptop screen. Video section: starting at 00:13.

Part E: Elastic magnification of objects on a typical smartphone screen. Video section: starting at 00:17.

Part F: One-point perspective magnification of objects on typical laptop and smartphone screens. starting at 00:21.

Part G: Elastic magnification of objects on typical laptop and smartphone screens. Video section: starting at 00:25.

Part H: One-point perspective magnification and Elastic magnification of objects on typical laptop screens. Video section: starting at 00:29.

Part I: One-point perspective magnification and Elastic magnification of objects on typical smartphone screens. Video section: starting at 00:34.

A.2 Video Fig. 2. Use Case Exploration Introduction: Linecept

This video introduces the Linecept use case through a hypothetical scenario. File: vidFig2.mp4 attached to the submitted manuscript.

A.3 Video Fig. 3. User Interfaces and Interactions Introduced in the Empirical Study

This video introduces all the user interfaces that were used for the empirical study both on a typical laptop as well as smartphone screens. File: vidFig3.mp4

Type 1: Modal type.

Type 2: Drawer type.

Type 3: Scale modal type.

Type 4: EZUI type.